

Column Comments Mission

Automated, standards-compliant column descriptions
for Data Lake tables

Audience: Data Engineers

Length: 20 min + Q&A

Repo: `gdcorp-dna/moon-unit-missions`

The problem

Every Data Lake table needs column-level descriptions. Today:

- 🐢 **Slow** — writing 50–200 comments per table, by hand
- 🎲 **Inconsistent** — `GCR` becomes "Gross Customer Receipt", "Gross Cash Receipts", or just `gcr`
- 📅 **Stale** — once written, comments rarely keep up with schema changes or business-term updates
- 🕵️ **Knowledge silos** — context lives in Confluence pages, Slack threads, and the head of one engineer
- 📏 **Hard limits get hit** — Hive's 256-byte COMMENT cap silently truncates anything past 255 chars

Result: column comments are either missing, wrong, or untrustworthy.

What this mission produces

A 3-stage Moon Units pipeline that:

1. **Researches** the table — DDL, Confluence design pages, Alation catalog, Certified Data Dictionary
2. **Enriches** the DDL with COMMENT clauses on every column, applying the GoDaddy Column Description Standard
3. **Validates** the 255-char limit and condenses overflowing comments

Output: a **production-ready** `table.ddl` you can PR straight into `gdcorp-dna/lake`, plus a complete audit trail.

Status today: 18 tables enriched across 9 schemas (`enterprise`, `customer360`, `ecomm360`, `analytic`, `gd_traffic_mart`, ...)

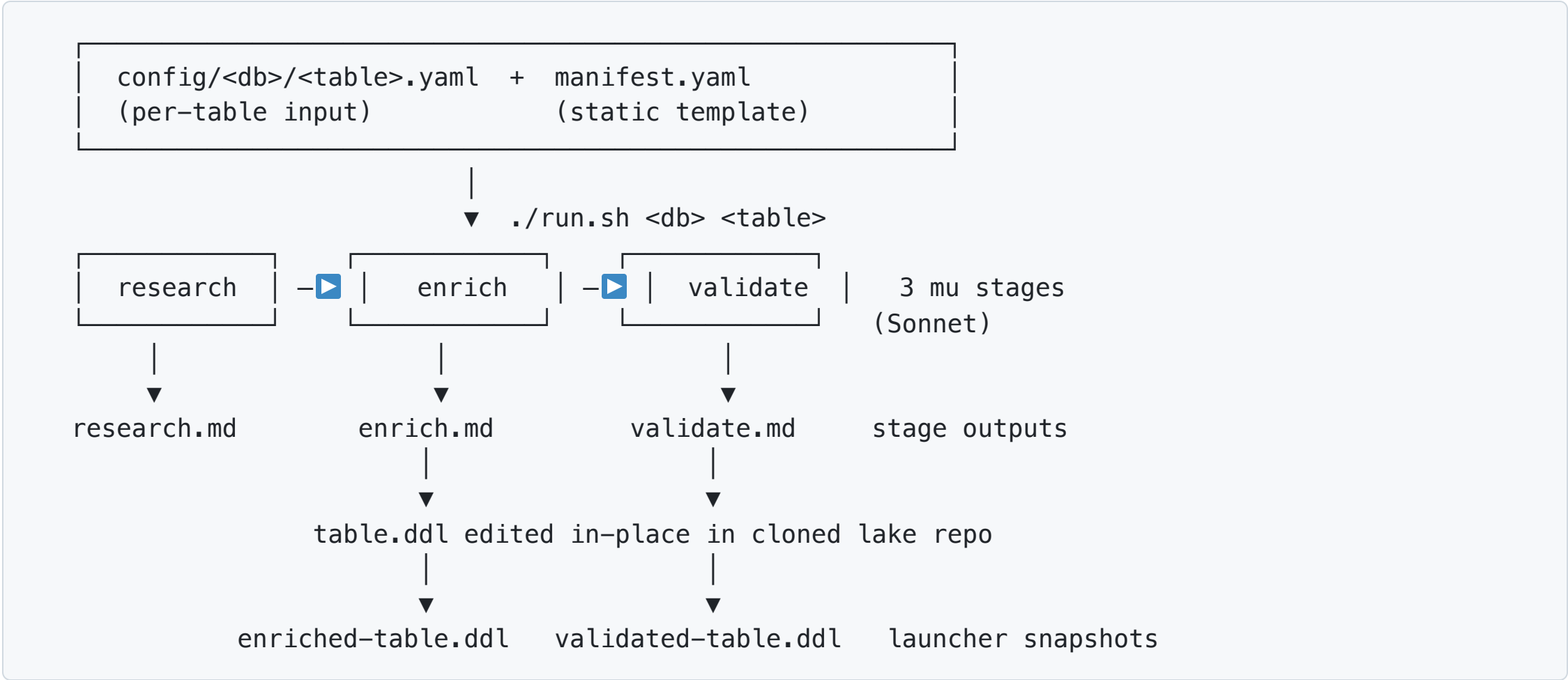
Before / After — `enterprise.fact_bill_line`

```
-- BEFORE (what's in gdcorp-dna/lake today, for many tables)
CREATE TABLE fact_bill_line(
  bill_id          string
  ,bill_line_num   int
  ,source_system_name string
  ,refund_flag     boolean
  ,gcr_usd_amt     decimal(18,2)
  -- 69 more columns, no comments
);
```

Before / After — `enterprise.fact_bill_line`

```
-- AFTER (what the mission produced – 74 columns, all annotated)
CREATE TABLE fact_bill_line(
  bill_id          string COMMENT '@PrimaryKey Unique identifier of the
    billing receipt (order). Maps to order_id in legacy e-commerce source
    tables or subscription_order_id for Smartline. Part of composite primary
    key (bill_id, bill_line_num, source_system_name).'
  ,bill_line_num    int      COMMENT '@PrimaryKey Line item number within a
    billing receipt. Together with bill_id and source_system_name forms the
    composite primary key.'
  ,source_system_name string COMMENT '@PrimaryKey Name of the source
    e-commerce system. @Enumerated(legacy e-comm, new e-comm, Smartline
    subscription store name).'
  ,refund_flag      boolean COMMENT 'Indicates whether this bill line is a
    refund transaction (true/false). True when bill_id contains the
    character R.'
  ,gcr_usd_amt      decimal(18,2) COMMENT 'Gross Cash Receipts (GCR) for
    this line, in US dollars. Sum of receipt_price minus refunds and
    chargebacks.'
);
```

The pipeline at a glance



Data & Analytics Column Comments
Three sources of truth feed research :

Stage 1: Research

Reads:

- The target `table.ddl` and `table.yaml` from `gdcorp-dna/lake` (cloned at start)
- All Confluence URLs listed in the per-table config
- Alation: target table metadata + columns (incl. existing `column_comment` Source Comments)
- Alation: reference tables' columns (predecessor / sibling tables)
- Certified Data Dictionary documents (paginated)

Produces `research.md` with:

- The full current DDL
- Confluence page summaries
- A **Certified Data Dictionary Mappings** table (mandatory)
- Per-column inferred purpose + context from every source

Stage 2: Enrich — applies the Column Description Standard

12 mandatory rules. The agent enforces them; we don't.

#	Rule	What it looks like
1	Be clear and concise	"Indicates whether..." not "This is the column that indicates whether..."
4	Avoid abbreviations	GCR → "Gross Cash Receipts (GCR)"
5	Indicate units & scale	...in US dollars , ...in milliseconds , ...(true/false)
8	Key annotations	@PrimaryKey , @ForeignKey(dest_table) , @Enumerated(v1, v2, ...)
10	At least one PK column	Every table must declare its primary key
11	Audit columns include TZ	etl_build_mst_ts documents Mountain Standard Time
12	Preserve 'Employee PII'	Never overwritten — appended to

Full standard: <https://godaddy-corp.atlassian.net/wiki/spaces/BI/pages/.../Column+Description+Standard>

Stage 2: Enrich — what the agent appends to `enrich.md`

Enrichment summary

- Target: `enterprise.fact_bill_line`
- Columns touched: 74
- Columns with pre-existing comments preserved: 3 ('Employee PII' annotations)
- Columns newly annotated: 71

Certified Data Dictionary terms applied

Abbreviation	Official expansion	Where used	
--- --- ---			
GCR	Gross Cash Receipts	<code>gcr_usd_amt</code> , <code>margin_gcr_usd_amt</code>	
MSRP	Manufacturer's Suggested	<code>original_list_price_usd_amt</code>	
	Retail Price		

Notable decisions

- Preserved the `'Employee PII'` annotation on `shopper-id` columns and appended descriptive text after it.
- Expanded GCR using the Certified Data Dictionary; rejected paraphrases like "Gross Customer Receipt" the agent had drafted.

Stage 3: Validate

The 255-char enforcement layer. Every COMMENT '...' is recounted.

If a comment exceeds 255 chars, it's condensed using rules **in order**:

1. Drop parenthetical synonyms / aliases
2. Drop verbose qualifiers ("that is used for" → remove)
3. Shorten @Enumerated lists (keep top 2-3 values, add "etc.")
4. Drop secondary context sentences
5. Tighten phrasing ("Indicates whether" → "Whether")
6. Last resort: drop least-essential clause

Never mid-word truncation. **Never** ...-ellipsis cutoff.

Output: validate.md reports condensed columns with before/after lengths.

How a data engineer uses it

Once per table:

```
cd missions/column-comments

# 1. Author the per-table config (use the helper scripts!)
#   config/enterprise/fact-bill-line.yaml – fills in:
#     – target.db_name / table_name / registry_path
#     – confluence_pages: [...]
#     – reference_tables: [...]
#     – alation: {enabled: true}

# 2. Run it
./run.sh enterprise fact-bill-line

# 3. Review output/enterprise/fact-bill-line/
#     validated-table.ddl      ← PR this into gdcorp-dna/lake
#     ddl-comparison.md       ← side-by-side diff
#     research.md / enrich.md / validate.md ← audit trail
```

Authoring configs is automated too

Don't hand-write configs. Use the `column-comments-config` Claude Code skill, which chains four helper scripts:

Script	Purpose
<code>alation_fetch_table_metadata.py</code>	Pull description + custom_fields from Alation; extract Confluence URLs
<code>fetch_alation_catalog_set_design.py</code>	Read Catalog Set's shared "Data Design" link from <code>/api/v1/table/<id>/shared_catalog_sets[].description</code>
<code>confluence_search_bi_space.py</code>	CQL search of the BI Confluence space, with noise filter and excerpts
<code>lake_lineage_fetch.py</code>	Pull <code>table.yaml</code> lineage; resolve upstream deps to Alation IDs

```
# In Claude Code:
> create a config for enterprise.dim_subscription
# (skill chains the scripts, populates the YAML, updates CATALOG.md)
```

Full playbook: `missions/column-comments/docs/creating-a-new-config.md`

Per-table config — what humans actually edit

```
# config/enterprise/fact-bill-line.yaml
target:
  db_name: "enterprise"
  table_name: "fact-bill-line"
  registry_path: "standard" # or "dlms-api"

confluence_pages:
- url: "https://godaddy-corp.atlassian.net/wiki/spaces/BI/pages/10371978/Fact_Bill_Line"
  description: "Fact_Bill_Line design specification"

reference_tables:
- name: "fact_bill_line_vw"
  schema: "ecomm360"
  alation_table_id: 7027689
  description: "Successor table – most column descriptions are relevant"

alation:
  enabled: true
  search_query: null # defaults to db_name.table_name
```

Why we trust the output

Trust mechanism	What it protects against
Three sources of truth	Single-source bias — Alation, Confluence, Dictionary cross-check each other
Certified Data Dictionary lookup	Hallucinated abbreviation expansions ("GCR ≠ Gross Customer Receipt")
Reference-table column carryover	Reinventing definitions when the predecessor table already documented them
PII annotation preservation	Stripping compliance-critical tags during enrichment
In-repo DDL edit + 3 snapshots	Loss of the <code>original-table.ddl</code> baseline for review
Stage-3 length validation	Silent DB-engine truncation past 255 chars
Per-stage <code>.md</code> audit trail	"Why did the agent write this?" — every decision is captured
<code>ddl-comparison.md</code>	Reviewers spotting unintended changes during PR

Audit trail per run

```
output/<db>/<table>/  
├─ INPUT.md           ← parameters the mission ran with  
├─ research.md        ← stage 1: Confluence + Alation + Dictionary  
├─ enrich.md          ← stage 2: enrichment summary + decisions  
├─ validate.md        ← stage 3: 255-char check + rewrites  
├─ original-table.ddl ← snapshot pre-enrich  
├─ enriched-table.ddl ← snapshot post-enrich  
├─ validated-table.ddl ← snapshot post-validate (authoritative)  
└─ ddl-comparison.md  ← per-column Original | Enriched | Validated | Len
```

All committed to the missions repo as a permanent record.

Reviewing a PR? Open `ddl-comparison.md` next to `validated-table.ddl`. Done.

What we've enriched (as of today)

18 tables across 9 schemas — all `enriched` status in `CATALOG.md` :

Bill / receipt facts

- `enterprise.fact_bill_line`
- `enterprise.fact_entitlement_bill`
- `bi_reports.ads_entitlement_bill`
- `analytic.ads_bill_line`
- `ecomm360.fact_bill_line_vw`
- `ecomm360.dim_bill_vw`
- `ecomm_mart.bill_line_traffic_ext`
- `ecomm_mart.renewal_360`

Subscription / entitlement

- `enterprise.dim_entitlement` (+ history)
- `enterprise.dim_subscription` (+ history)

Lessons we paid for so we don't have to again

- **Alation's Catalog Set Description** doesn't live where the docs say. It's at `/api/v1/table/<id>/` under `shared_catalog_sets[].description`, not at `/integration/v2/custom_field_value/?otype=dynamic_set_property` (which only returns Title for catalog sets).
- **Lake registry uses hyphens, Alation/SQL use underscores.** Configs split this; tools take the form they expect.
- `mu launch --keep-container` outlives `mu launch` — workspace cleanup must `docker stop` the container first, not just SIGTERM the launcher.
- **COMMENT casing is non-uniform** — agents emit both `COMMENT` and `comment`; downstream parsing must be case-insensitive.
- **Stage output:** is markdown, not arbitrary file. The framework wraps it; agents append, don't overwrite.

Limitations & roadmap

Today:

- Sonnet-only (cost: ~\$0.50–\$1.50 per table run)
- Per-table sequential runs; no batch mode
- Manual PR step from `validated-table.ddl` → `gdcorp-dna/lake`
- Confluence search noise filter is hand-tuned; may miss design pages with unusual titles

Considering:

- Batch run-all- `stale` -tables driven by `CATALOG.md`
- Re-enrich on schema change (detect via `git log` on the DDL file)
- Auto-PR back to `gdcorp-dna/lake` (gated by CI lint)
- Extend to column descriptions in Tableau / Looker downstream

Not on roadmap:

Try it

```
git clone https://github.com/jxhuang-godaddy/moon-unit-missions
cd moon-unit-missions/missions/column-comments
cp .env.local.example .env.local          # fill in tokens
./run.sh                                  # lists configurable tables
./run.sh enterprise fact-bill-line        # actual run
```

Repo paths to read:

- `README.md` — usage
- `docs/data-flow.md` — mermaid diagrams of every stage
- `docs/creating-a-new-config.md` — config-authoring playbook
- `CLAUDE.md` — gotchas & API quirks (the war stories)

Owner: @mzwolak · **Built with:** Moon Units, Claude Sonnet 4.6, Alation API, Confluence API

Questions?

Slides built with Marp · `missions/column-comments/docs/demo-slides.md`